

MECKLENBURG-BRUNSWICK, VA, USA

WMO: 724118

Lat: 36.688N

Lon: 78.054W

Elev: 442

StdP: 14.46

Time Zone: -5.00 (NAE)

Period: 97-19

WBAN: 93797

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	18.0	21.8	1.0	5.9	23.1	6.8	7.9	29.0	19.4	54.8	17.3	52.3	3.1	10	0.384

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.1	95.3	76.0	92.6	75.4	90.4	75.1	80.5	87.4	79.2	86.6	77.9	85.8	5.9	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	153.2	83.8	77.1	143.6	82.7	75.4	135.2	80.6	44.5	88.1	42.8	87.1	41.6	85.5	86.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
16.6	13.7	11.8	DB	8.7	98.6	6.7	4.2	3.8	101.6	-0.1	104.1	-3.8	106.4	-8.7	109.4
			WB	8.5	81.5	5.9	2.2	4.3	83.1	0.9	84.4	-2.4	85.6	-6.7	87.2

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	59.9	40.0	43.2	49.5	59.0	67.4	75.3	78.9	77.4	71.7	60.9	50.4	43.8
	DBStd	15.83	10.87	9.44	9.85	8.46	7.04	5.80	4.59	4.76	5.97	8.42	8.59	9.43
	HDD50	1077	346	232	134	17	0	0	0	0	0	10	97	241
	HDD65	3457	776	613	489	219	60	4	0	0	15	182	441	658
	CDD50	4686	36	40	118	286	540	760	896	850	651	349	110	50
	CDD65	1591	1	1	8	39	135	314	431	385	216	55	4	2
	CDH74	14974	3	6	91	434	1254	3096	4310	3594	1687	455	33	11
	CDH80	6213	0	0	12	109	401	1361	2034	1577	598	118	2	1
Wind	WSAvg	4.1	4.7	4.9	5.2	5.3	4.2	3.6	3.5	2.9	3.3	3.6	4.0	4.1
Precipitation	PrecAvg	43.8	3.3	2.6	4.3	3.3	3.7	3.9	4.6	4.1	4.1	3.3	3.0	3.3
	PrecMax	65.3	6.4	5.5	7.3	5.7	8.4	8.9	9.1	9.6	14.4	8.6	8.5	7.9
	PrecMin	32.6	0.9	1.0	0.5	1.1	1.3	1.4	2.0	1.6	0.3	0.0	0.4	0.7
	PrecStd	7.0	1.3	1.2	1.7	1.5	1.7	2.0	1.7	1.8	3.1	2.0	1.9	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.4	73.7	81.6	88.2	91.0	99.1	99.7	98.8	93.1	89.5	79.1	74.6
		MCWB	63.2	62.6	63.5	65.2	73.2	73.8	76.3	76.4	73.6	73.3	66.4	64.1
	2%	DB	66.4	69.5	75.4	82.3	88.1	93.2	95.5	93.4	90.2	82.3	73.0	68.5
		MCWB	60.9	58.3	60.9	64.9	71.7	74.4	76.6	76.6	73.4	70.1	63.3	62.2
	5%	DB	63.2	64.2	72.0	79.4	84.2	90.5	92.7	90.8	86.6	79.3	70.0	63.9
		MCWB	57.0	55.0	58.9	63.9	70.9	74.2	75.9	76.5	73.2	68.4	62.2	58.6
	10%	DB	57.3	60.6	66.3	74.8	81.5	88.1	90.3	88.3	82.7	74.9	65.5	60.5
		MCWB	50.5	52.5	56.5	62.0	68.6	73.8	75.9	75.9	71.8	66.6	59.5	54.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	66.4	65.7	66.6	71.8	76.9	79.8	82.3	82.6	80.9	76.9	71.5	68.1
		MCDB	69.3	70.0	75.4	78.9	85.0	87.6	89.6	90.4	83.8	82.1	73.3	70.7
	2%	WB	62.7	62.1	63.9	69.1	74.6	78.1	80.6	80.5	78.0	74.2	66.9	64.0
		MCDB	65.5	66.4	71.2	77.5	82.9	86.9	88.2	87.4	83.9	78.4	71.1	66.4
	5%	WB	58.2	58.0	61.8	66.6	72.8	76.9	79.3	79.1	76.2	72.0	63.8	60.0
		MCDB	62.1	63.3	68.7	74.9	80.9	85.7	87.3	86.2	81.3	76.8	66.5	63.8
	10%	WB	52.1	52.5	58.9	64.4	71.1	75.5	78.0	77.6	74.4	68.9	60.0	55.2
		MCDB	56.6	58.4	65.9	71.2	78.6	83.5	86.5	85.0	78.8	73.6	65.1	59.0
Mean Daily Temperature Range	5% DB	MDBR	19.4	20.9	22.5	23.4	21.8	21.0	20.1	19.6	19.7	21.6	21.8	19.1
		MCDBR	24.6	26.6	27.3	27.8	25.1	24.4	23.8	22.6	23.6	24.5	25.2	23.2
		MCWBR	18.4	18.3	15.1	13.2	11.5	9.1	8.8	8.9	10.0	13.2	16.2	17.4
	5% WB	MCDBR	22.0	23.8	23.8	22.7	20.8	20.2	19.4	18.7	17.7	18.9	20.0	20.2
		MCWBR	18.9	19.1	15.9	12.9	10.4	9.1	8.6	8.8	9.3	11.4	15.7	17.4
Clear-Sky Solar Irradiance	taub	0.298	0.312	0.343	0.386	0.432	0.478	0.508	0.510	0.401	0.355	0.317	0.300	
	taud	2.538	2.497	2.446	2.330	2.223	2.144	2.082	2.058	2.358	2.473	2.538	2.566	
	Ebn at Noon	287	295	293	284	271	258	249	246	270	275	277	278	
	Edh at Noon	25	29	33	40	45	49	51	51	36	30	24	23	
All-Sky Solar Radiation	RadAvg	776	1005	1320	1671	1864	2022	1906	1726	1414	1133	858	678	
	RadStd	58	117	100	132	151	135	101	91	131	140	69	58	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (63 neighbors)	+0.72	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+259	+173

Nomenclature: See separate page